ExoHabitAI: Dataset Documentation & Analytical Report

Project Title: ExoHabitAI – Machine Learning for Exoplanet Habitability Prediction
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1. Introduction

The search for habitable planets beyond our solar system has become one of the most exciting areas of modern astronomy. The ExoHabitAI project focuses on leveraging Machine Learning techniques to predict whether an exoplanet can potentially support life based on its physical and stellar properties.

This report provides a detailed explanation of the dataset used in the project, including its source, structure, preprocessing techniques, and feature engineering strategies. The raw astronomical data has been transformed into a refined and structured format suitable for building predictive models. Additionally, scientifically meaningful metrics such as the Earth Similarity Index (ESI) have been incorporated to enhance the evaluation of habitability.

2. Dataset Overview

The dataset for this project has been obtained from the **NASA Exoplanet Archive**, specifically the *Planetary Systems Composite Data Table*. This archive is widely recognized as one of the most reliable and comprehensive sources of confirmed exoplanet discoveries.

Dataset Summary

- **Source:** NASA Exoplanet Archive
- **Initial Size:** 39,386 observations with 289 features
- **Filtered Dataset:** 6,107 unique exoplanets
- **Primary Detection Methods:** Transit and Radial Velocity

The dataset originally contained multiple entries for the same planet due to updates and different observation methods. To ensure accuracy and consistency, the dataset was filtered using the `default_flag = 1` condition. This step helped retain only the most reliable and up-to-date record for each exoplanet, thereby reducing redundancy and improving data quality.

3. Feature Selection and Data Dictionary

Although the raw dataset contains a large number of attributes, only a subset of features was selected for model training. These features were chosen based on their direct relevance to planetary habitability and their availability across observations.

Planetary Characteristics

- **Planet Name:** A unique identifier assigned to each exoplanet.
- **Planet Radius:** Measured in Earth radii, this parameter helps determine whether the planet is rocky or gaseous. Smaller radii generally indicate rocky planets, which are more likely to support life.
- **Planet Mass:** Measured in Earth masses, it is crucial for understanding gravitational properties and estimating planetary density.
- **Orbital Period:** Represents the duration required for a planet to complete one full orbit around its host star.
- **Semi-Major Axis:** Indicates the average distance between the planet and its star, which directly affects temperature and radiation levels.

Stellar and Environmental Factors

- **Equilibrium Temperature:** An estimate of the planet's surface temperature, assuming no atmosphere. This plays a key role in determining whether liquid water can exist.
- **Stellar Effective Temperature:** The surface temperature of the host star, influencing the energy received by the planet.
- **Stellar Radius:** The size of the host star, which contributes to luminosity and radiation output.

4. Data Quality Assessment and Missing Value Handling

Astronomical datasets are inherently incomplete due to limitations in observation techniques and instruments. Therefore, a detailed analysis was conducted to identify missing values and determine appropriate strategies for handling them.

Missing Data Analysis

- **High Missing Values:**
  - Insolation Flux (approximately 85%)
  - Planetary Density (approximately 81%)
- **Moderate Missing Values:**
  - Planet Mass (around 50%), especially for planets detected using the transit method

Simply removing rows with missing values would significantly reduce the dataset size and eliminate potentially useful information. Therefore, a more sophisticated approach was required.

Physics-Based Imputation Strategy

Instead of using basic statistical imputation methods, scientifically grounded approaches were adopted:

- **Stefan-Boltzmann Law:** Used to estimate stellar luminosity and radiation flux when stellar radius and temperature were available.
- **Mass-Radius Relationships:** Applied to approximate missing mass values for rocky planets based on known physical correlations.

This method ensured that the imputed values remained realistic and consistent with astrophysical principles, thereby improving the reliability of the dataset.

5. Statistical Analysis and Observations

The dataset represents a diverse range of exoplanets, each with unique physical and environmental characteristics. Statistical analysis provides insight into the distribution and variability of these parameters.

Parameter Ranges

- **Planet Radius:** 0.3 to 87.2 Earth radii
- **Orbital Period:** From a few hours to more than 1000 years
- **Equilibrium Temperature:** 50 K to 4050 K

These values indicate the presence of a wide spectrum of planetary types, including small rocky planets, massive gas giants, extremely hot planets, and frozen worlds. Such diversity highlights the complexity of predicting habitability.

6. Target Variable Engineering (Habitability Classification)

The original dataset does not include a direct indicator of whether a planet is habitable. Therefore, a custom target variable was defined to enable supervised machine learning.

Criteria for Habitability

A planet is classified as potentially habitable if it satisfies the following conditions:

1. **Composition:** The planet must be rocky, which is typically indicated by a smaller radius or moderate mass.
2. **Temperature Range:** The equilibrium temperature should fall within a range that allows liquid water to exist.
3. **Stellar Energy Input:** The planet must receive a stable and suitable amount of radiation from its host star.

Earth Similarity Index (ESI)

To provide a more detailed ranking of planets, the Earth Similarity Index (ESI) was implemented. This index compares planetary characteristics such as radius, density, and temperature with Earth's values, producing a score that reflects how similar a planet is to Earth.

This metric allows for a more nuanced understanding of habitability rather than a simple binary classification.

7. Discovery Methods and Dataset Bias

The method used to detect exoplanets plays a significant role in shaping the dataset. Each detection technique has its own strengths and limitations, which can introduce bias.

- The **Transit Method** detects planets when they pass in front of their host star, causing a slight dip in brightness. This method is more sensitive to larger planets close to their stars.
- The **Radial Velocity Method** measures variations in a star's motion caused by the gravitational pull of orbiting planets.

Since most planets in the dataset were discovered using the transit method, there is a bias toward detecting larger and closer planets, which may not necessarily represent the full diversity of planetary systems.

8. Conclusion and Future Scope

The dataset has been successfully cleaned, processed, and transformed into a machine learning-ready format. By reducing redundancy, handling missing values using scientific methods, and engineering meaningful features, a high-quality dataset of 6,107 exoplanets has been prepared.

The next phase of the project involves training machine learning models such as **Random Forest** and **XGBoost** to predict planetary habitability. These models will help identify potential candidates for life beyond Earth and contribute to ongoing research in space exploration and astrobiology.